

Principal Component Analysis (PCA) on Boston Housing Dataset

Set Working Directory

```
setwd("/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/descriptive_analysis")
getwd()
```

```
## [1] "/Users/Ian/Documents/Code_Projects/boston-house-price-modeling/descriptive_analysis"
```

1. Load Libraries

```
library(ggplot2)
library(dplyr)
library(ggrepel)  # for non-overlapping labels
library(patchwork)
```

2. Load Dataset

```
# Read the cleaned dataset (assumes 'B' column removed)
data <- read.csv("../data/boston_cleaned.csv")
str(data)
```

```
## 'data.frame':  506 obs. of  13 variables:
## $ CRIM   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ ZN     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ INDUS  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ CHAS   : int   0 0 0 0 0 0 0 0 0 0 ...
## $ NOX    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ RM     : num  6.58 6.42 7.18 7 7.15 ...
## $ AGE    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ DIS    : num  4.09 4.97 4.97 6.06 6.06 ...
## $ RAD    : int   1 2 2 3 3 3 5 5 5 5 ...
## $ TAX    : int  296 242 242 222 222 222 311 311 311 311 ...
## $ PTRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ LSTAT  : num  4.98 9.14 4.03 2.94 5.33 ...
## $ MEDV   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

3. Basic Statistics and Correlation

```
# Correlation matrix
cor(data)
```

```
##          CRIM          ZN          INDUS          CHAS          NOX
## CRIM      1.00000000 -0.20046922  0.40658341 -0.055891582  0.42097171
## ZN        -0.20046922  1.00000000 -0.53382819 -0.042696719 -0.51660371
## INDUS     0.40658341 -0.53382819  1.00000000  0.062938027  0.76365145
## CHAS     -0.05589158 -0.04269672  0.06293803  1.000000000  0.09120281
## NOX       0.42097171 -0.51660371  0.76365145  0.091202807  1.00000000
## RM       -0.21924670  0.31199059 -0.39167585  0.091251225 -0.30218819
## AGE       0.35273425 -0.56953734  0.64477851  0.086517774  0.73147010
## DIS      -0.37967009  0.66440822 -0.70802699 -0.099175780 -0.76923011
## RAD       0.62550515 -0.31194783  0.59512927 -0.007368241  0.61144056
## TAX       0.58276431 -0.31456332  0.72076018 -0.035586518  0.66802320
## PTRATIO  0.28994558 -0.39167855  0.38324756 -0.121515174  0.18893268
## LSTAT    0.45562148 -0.41299457  0.60379972 -0.053929298  0.59087892
## MEDV    -0.38830461  0.36044534 -0.48372516  0.175260177 -0.42732077
##          RM          AGE          DIS          RAD          TAX          PTRATIO
## CRIM     -0.21924670  0.35273425 -0.37967009  0.625505145  0.58276431  0.28994558
## ZN       0.31199059 -0.56953734  0.66440822 -0.311947826 -0.31456332 -0.39167855
## INDUS    -0.39167585  0.64477851 -0.70802699  0.595129275  0.72076018  0.38324756
## CHAS     0.09125123  0.08651777 -0.09917578 -0.007368241 -0.03558652 -0.1215152
## NOX     -0.30218819  0.73147010 -0.76923011  0.611440563  0.66802320  0.1889327
## RM       1.00000000 -0.24026493  0.20524621 -0.209846668 -0.29204783 -0.3555015
## AGE     -0.24026493  1.00000000 -0.74788054  0.456022452  0.50645559  0.2615150
## DIS     0.20524621 -0.74788054  1.00000000 -0.494587930 -0.53443158 -0.2324705
## RAD     -0.20984667  0.45602245 -0.49458793  1.000000000  0.91022819  0.4647412
## TAX     -0.29204783  0.50645559 -0.53443158  0.910228189  1.00000000  0.4608530
## PTRATIO -0.35550149  0.26151501 -0.23247054  0.464741179  0.46085304  1.0000000
## LSTAT   -0.61380827  0.60233853 -0.49699583  0.488676335  0.54399341  0.3740443
## MEDV    0.69535995 -0.37695457  0.24992873 -0.381626231 -0.46853593 -0.5077867
##          LSTAT          MEDV
## CRIM      0.4556215 -0.3883046
## ZN        -0.4129946  0.3604453
## INDUS     0.6037997 -0.4837252
## CHAS     -0.0539293  0.1752602
## NOX       0.5908789 -0.4273208
## RM       -0.6138083  0.6953599
## AGE       0.6023385 -0.3769546
## DIS      -0.4969958  0.2499287
## RAD       0.4886763 -0.3816262
## TAX       0.5439934 -0.4685359
## PTRATIO  0.3740443 -0.5077867
## LSTAT     1.0000000 -0.7376627
## MEDV     -0.7376627  1.0000000
```

4. PCA Analysis

```
# Run PCA with scaling
pr.out <- prcomp(data, scale. = TRUE)
```

4.1 PCA Object Details

```
# Structure of the PCA object  
str(pr.out)
```

```
## List of 5  
## $ sdev      : num [1:13] 2.515 1.284 1.123 0.939 0.901 ...  
## $ rotation: num [1:13, 1:13] 0.2428 -0.2544 0.3392 -0.0034 0.3313 ...  
## ..- attr(*, "dimnames")=List of 2  
## .. ..$ : chr [1:13] "CRIM" "ZN" "INDUS" "CHAS" ...  
## .. ..$ : chr [1:13] "PC1" "PC2" "PC3" "PC4" ...  
## $ center   : Named num [1:13] 3.6135 11.3636 11.1368 0.0692 0.5547 ...  
## ..- attr(*, "names")= chr [1:13] "CRIM" "ZN" "INDUS" "CHAS" ...  
## $ scale    : Named num [1:13] 8.602 23.322 6.86 0.254 0.116 ...  
## ..- attr(*, "names")= chr [1:13] "CRIM" "ZN" "INDUS" "CHAS" ...  
## $ x        : num [1:506, 1:13] -2.04 -1.3 -2.34 -2.8 -2.73 ...  
## ..- attr(*, "dimnames")=List of 2  
## .. ..$ : NULL  
## .. ..$ : chr [1:13] "PC1" "PC2" "PC3" "PC4" ...  
## - attr(*, "class")= chr "prcomp"
```

```
# Means and standard deviations  
pr.out$center
```

```
##          CRIM          ZN          INDUS          CHAS          NOX          RM  
## 3.61352356 11.36363636 11.13677866 0.06916996 0.55469506 6.28463439  
##          AGE          DIS          RAD          TAX          PTRATIO          LSTAT  
## 68.57490119 3.79504269 9.54940711 408.23715415 18.45553360 12.65306324  
##          MEDV  
## 22.53280632
```

```
pr.out$scale
```

```
##          CRIM          ZN          INDUS          CHAS          NOX          RM  
## 8.6015451 23.3224530 6.8603529 0.2539940 0.1158777 0.7026171  
##          AGE          DIS          RAD          TAX          PTRATIO          LSTAT  
## 28.1488614 2.1057101 8.7072594 168.5371161 2.1649455 7.1410615  
##          MEDV  
## 9.1971041
```

4.2 Loadings (Eigenvectors)

```
# Loadings of original variables on PCs  
pr.out$rotation
```

```
##          PC1          PC2          PC3          PC4          PC5  
## CRIM      0.242755751 -0.06016230 0.43465171 -0.1260139870 0.176504934  
## ZN        -0.254412844 -0.13972249 0.38597315 -0.3216822038 0.352165441  
## INDUS     0.339194411 0.12419215 -0.04164821 -0.0023181436 0.002899489
```

```
## CHAS    -0.003398861  0.40854248 -0.14313930 -0.8161803152 -0.362235049
## NOX      0.331332714  0.25241714 -0.04599276  0.0004143038  0.243834048
## RM       -0.210573420  0.44132080  0.34461081  0.2732487649 -0.100167568
## AGE      0.305261232  0.25534756 -0.19463553  0.1216651255  0.122383625
## DIS      -0.305937761 -0.35492911  0.14762383 -0.2112788801 -0.037703558
## RAD      0.305497402  0.03644414  0.48217440 -0.0490273484 -0.116204455
## TAX      0.327461336  0.01271806  0.40399465 -0.0519315778 -0.040103991
## PTRATIO  0.213780264 -0.31735473  0.08440322  0.1461790348 -0.736675882
## LSTAT    0.317265523 -0.20463716 -0.16924882 -0.1778127584  0.254960026
## MEDV     -0.271026715  0.44809782  0.16690885  0.1493079709 -0.070051426
##          PC6          PC7          PC8          PC9          PC10         PC11
## CRIM     -0.75709273  0.17327568 -0.25295388  0.09843268  0.07944524  0.06870346
## ZN        0.29580957 -0.36574375 -0.44610637 -0.15966967  0.10553619 -0.23496080
## INDUS     0.34757621  0.17395211 -0.54468250  0.40287510 -0.25828389  0.35886650
## CHAS     -0.08427406 -0.03774064  0.03480146  0.04410580  0.01032496 -0.01971422
## NOX       0.20576835  0.04738903  0.08108274  0.17192327  0.44388254 -0.44417536
## RM       -0.09055120 -0.45176103  0.14282252  0.52425472 -0.19526717 -0.09981588
## AGE      -0.11337958 -0.57759429 -0.05184518 -0.29908009  0.32918300  0.47098405
## DIS       0.10052574 -0.13367421  0.22732196  0.21629959  0.14860221  0.50807134
## RAD       0.14871263  0.07819568  0.41178723 -0.23767501 -0.03947623  0.02270508
## TAX       0.32937655  0.07638123  0.16024377 -0.05821442 -0.03153369  0.17784515
## PTRATIO -0.01244447 -0.28205841 -0.30394090 -0.08424491  0.08129786 -0.24251699
## LSTAT    -0.08183292 -0.35758793  0.16089478 -0.08038309 -0.68327427 -0.15935846
## MEDV      0.02096725  0.16532065 -0.21038869 -0.53524847 -0.27221781  0.08560370
##          PC12         PC13
## CRIM      0.09935538 -0.06085059
## ZN        -0.13364822  0.09640080
## INDUS      0.08242517  0.23543477
## CHAS     -0.05009232 -0.02325216
## NOX       0.53101848 -0.08894508
## RM       -0.04108085 -0.01143703
## AGE      -0.05916301  0.04068742
## DIS       0.55120464 -0.04575631
## RAD      -0.00531307  0.63289303
## TAX      -0.23933480 -0.70128170
## PTRATIO  0.18676090 -0.05392456
## LSTAT     0.25632034 -0.08559070
## MEDV      0.46268704 -0.12907177
```

4.3 Principal Component Scores

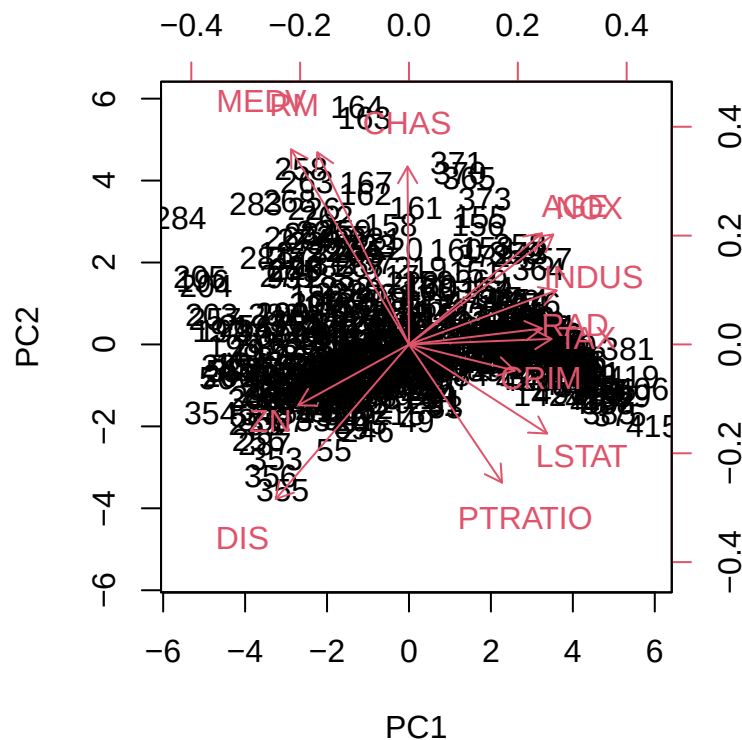
```
# Principal component scores (rows = observations, cols = PCs)
head(pr.out$x)
```

```
##          PC1          PC2          PC3          PC4          PC5          PC6
## [1,] -2.037670  0.4897836 -0.44383937  0.3397825  0.95404730 -0.3179268
## [2,] -1.302803 -0.1894539 -0.96969949  0.5344734 -0.07906241 -0.5788696
## [3,] -2.337875  0.9136400 -0.11305558  1.0945668 -0.54759373 -0.5172111
## [4,] -2.804963  0.1989315  0.05686570  0.9147103 -0.97602062 -0.6645884
## [5,] -2.733818  0.4363943  0.06788642  0.9943705 -0.89598462 -0.7418584
## [6,] -2.255556 -0.3348388 -0.45014059  0.6167837 -0.72216547 -0.6798518
##          PC7          PC8          PC9          PC10          PC11          PC12
```

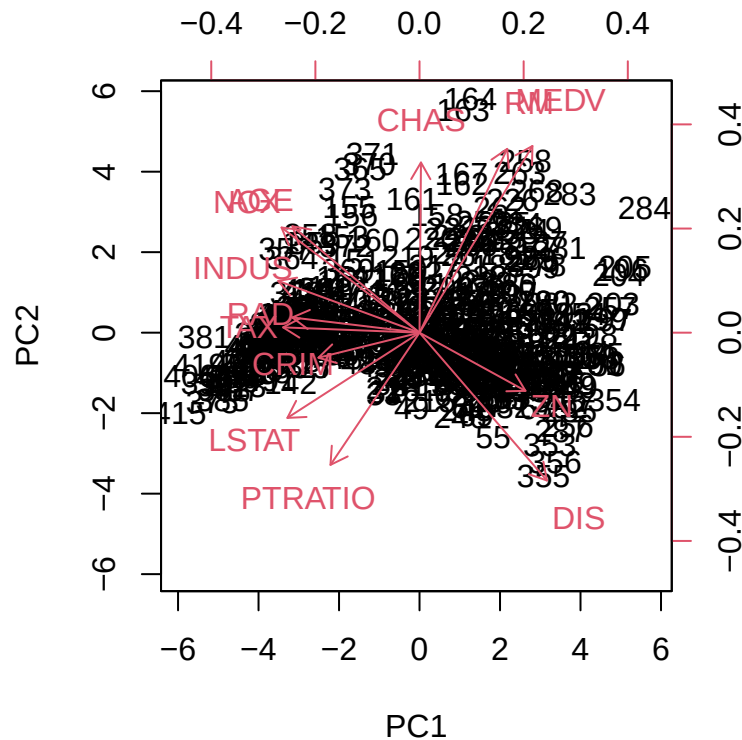
```
## [1,]  0.16080289 0.481402981  0.03687385  0.794933457 -0.1169477 -0.4905034
## [2,] -0.29445238 0.230631001  0.15405232  0.307510961  0.5911804 -0.1159977
## [3,]  0.07092113 0.003913368  0.20836463 -0.011774418  0.4207806  0.3523601
## [4,]  0.22106901 0.399804741  0.07297468  0.256343428  0.1348944  0.5739160
## [5,] -0.11570458 0.403339736 -0.09453535  0.001949619  0.2272947  0.7746213
## [6,]  0.12337205 0.419320987 -0.23995058  0.486945026  0.3370046  0.4250157
##
##      PC13
## [1,] -0.25126325
## [2,]  0.11268488
## [3,] -0.04807782
## [4,] -0.08555679
## [5,] -0.14404091
## [6,] -0.01889502
```

5. Visualize PCA: Biplot

```
# Basic biplot
biplot(pr.out, scale = 0)
```



```
# Optional: Flip PC1 direction (no impact on interpretation)
pr.out$rotation[,1] = -pr.out$rotation[,1]
pr.out$x[,1] = -pr.out$x[,1]
biplot(pr.out, scale = 0)
```



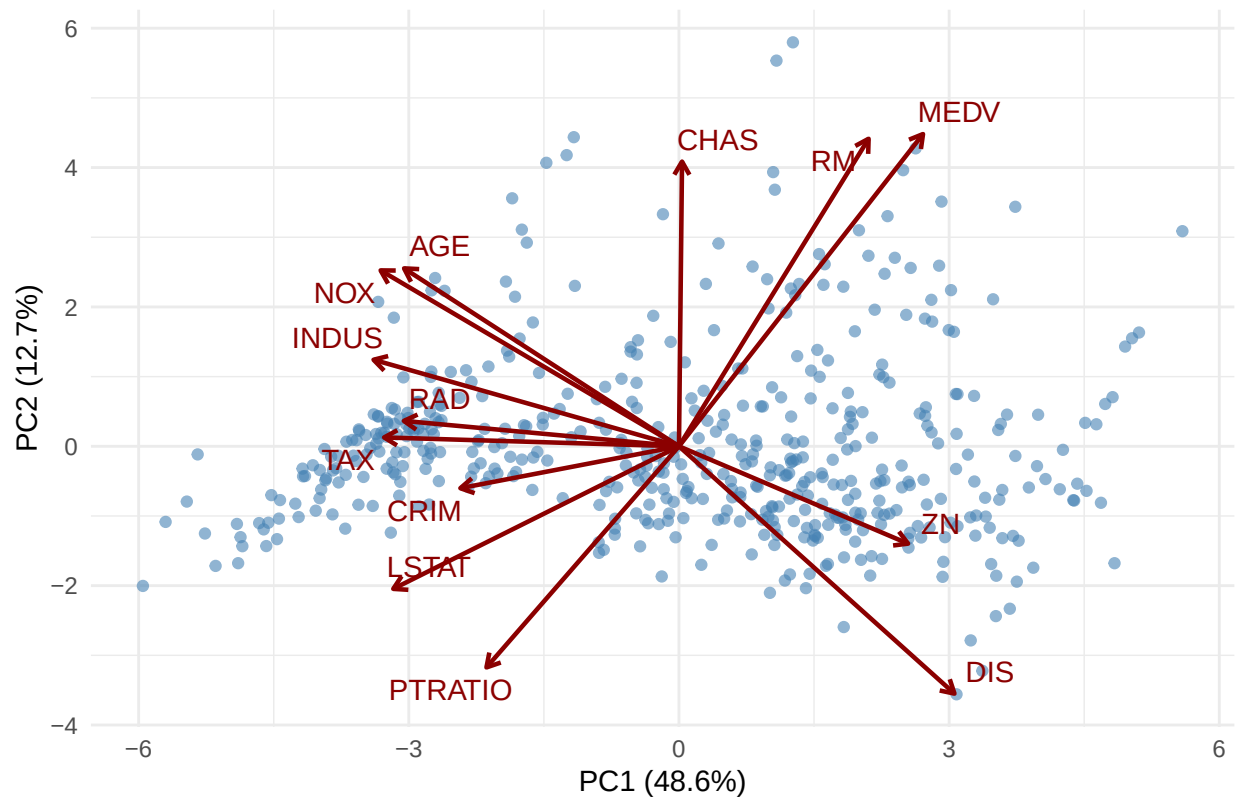
```
# Extract scores and loadings
scores <- as.data.frame(pr.out$x[, 1:2]) # PC1 and PC2
loadings <- as.data.frame(pr.out$rotation[, 1:2])
loadings$varname <- rownames(loadings)

# Scale loadings for visualization (arrows)
loadings_scaled <- loadings
loadings_scaled[, 1:2] <- loadings_scaled[, 1:2] * 10 # adjust scale for better arrow visibility

# Create biplot
p <- ggplot(scores, aes(x = PC1, y = PC2)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_segment(data = loadings_scaled,
    aes(x = 0, y = 0, xend = PC1, yend = PC2),
    arrow = arrow(length = unit(0.2, "cm")),
    color = "darkred", linewidth = 0.8) +
  geom_text_repel(data = loadings_scaled,
    aes(x = PC1, y = PC2, label = varname),
    color = "darkred", size = 4) +
  labs(title = "PCA Biplot (PC1 vs PC2)",
    x = paste0("PC1 (", round(summary(pr.out)$importance[2, 1] * 100, 1), "%)"),
    y = paste0("PC2 (", round(summary(pr.out)$importance[2, 2] * 100, 1), "%)")) +
  theme_minimal()

# Show plot
print(p)
```

PCA Biplot (PC1 vs PC2)



```
# Save to file
ggsave(filename = "../output/pca/pca_biplot.png", plot = p, width = 8, height = 8, dpi = 300, bg="white")
```

6. Variance Explained

```
# Summary of PCA: includes proportion of variance explained
pca_summary <- summary(pr.out)
pca_summary
```

```
## Importance of components:
##              PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.5146 1.2841 1.12316 0.93851 0.90053 0.73240 0.63544
## Proportion of Variance 0.4864 0.1268 0.09704 0.06775 0.06238 0.04126 0.03106
## Cumulative Proportion 0.4864 0.6133 0.71030 0.77805 0.84043 0.88170 0.91276
##              PC8    PC9    PC10    PC11    PC12    PC13
## Standard deviation  0.52930 0.51187 0.46200 0.42873 0.36641 0.24596
## Proportion of Variance 0.02155 0.02015 0.01642 0.01414 0.01033 0.00465
## Cumulative Proportion 0.93431 0.95446 0.97088 0.98502 0.99535 1.00000
```

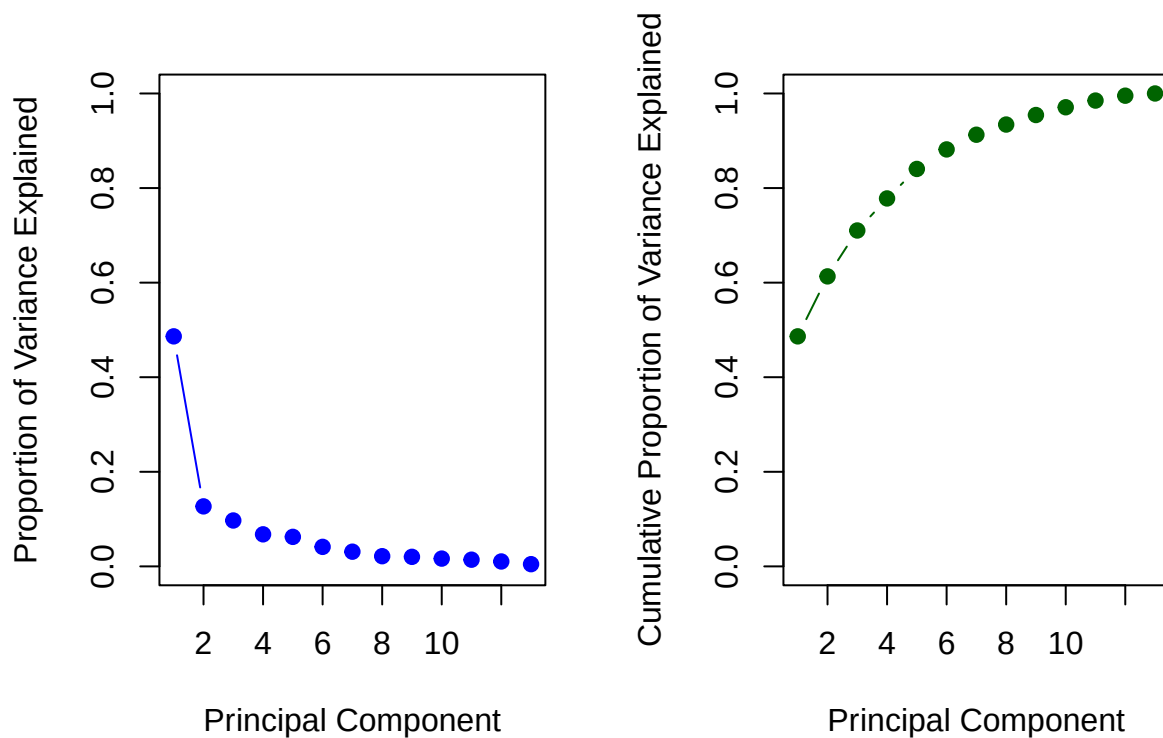
7. Scree Plot & Cumulative Variance

a. Plot

```
# Extract importance matrix
importance <- pca_summary$importance

# Scree plot and cumulative variance
par(mfrow = c(1, 2))
plot(importance[2,], xlab = "Principal Component",
     ylab = "Proportion of Variance Explained", ylim = c(0, 1),
     type = "b", col = "blue", pch = 19)

plot(importance[3,], xlab = "Principal Component",
     ylab = "Cumulative Proportion of Variance Explained",
     ylim = c(0, 1), type = "b", col = "darkgreen", pch = 19)
```



```
par(mfrow = c(1, 1))
```


b. ggplot2 Version

```
# Extract variance explained from PCA summary
pca_summary <- summary(pr.out)
importance <- pca_summary$importance

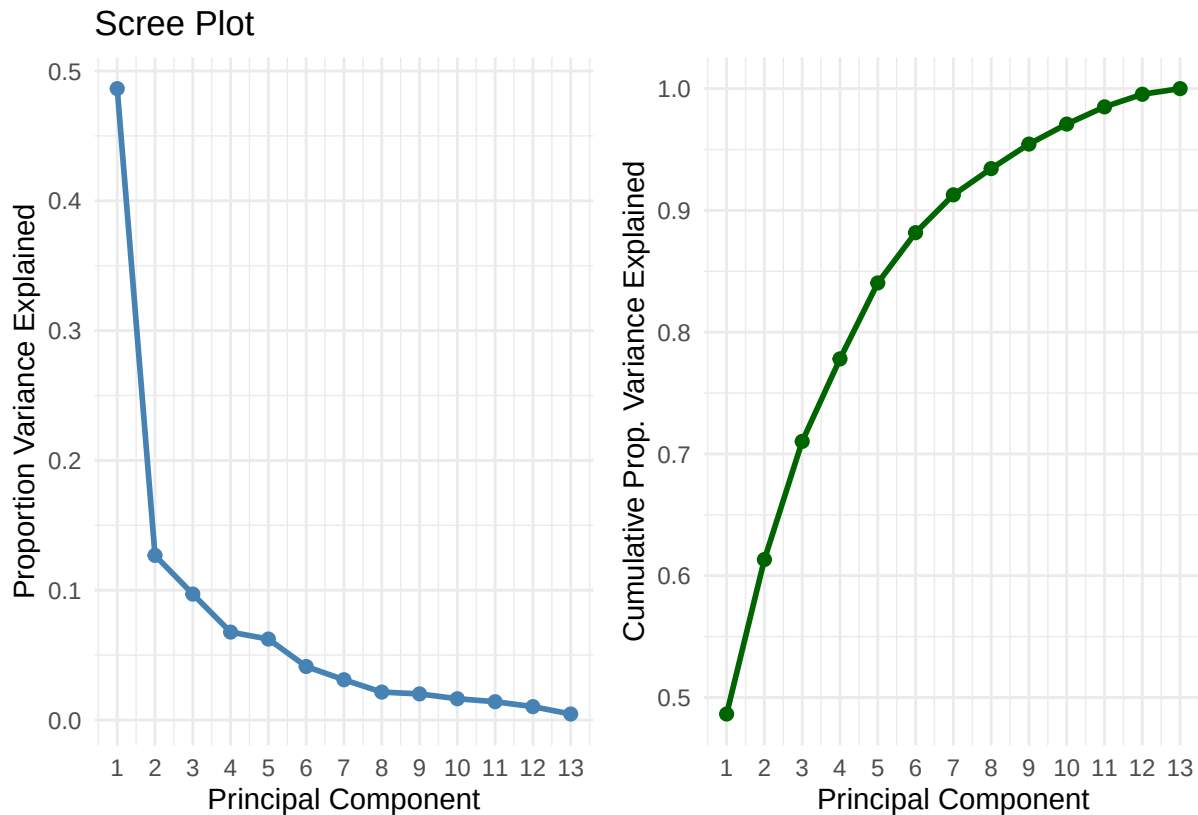
# Create a data frame for plotting
pc_data <- data.frame(
  PC = paste0("PC", 1:ncol(pr.out$rotation)),
  Variance = importance[2, ],
  Cumulative = importance[3, ],
  PC_num = 1:ncol(pr.out$rotation)
)

# Scree Plot
scree_plot <- ggplot(pc_data, aes(x = PC_num, y = Variance)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "steelblue", size = 2) +
  scale_x_continuous(breaks = pc_data$PC_num) +
  labs(title = "Scree Plot",
       x = "Principal Component",
       y = "Proportion Variance Explained") +
  theme_minimal()

# Cumulative Variance Plot
cum_plot <- ggplot(pc_data, aes(x = PC_num, y = Cumulative)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "darkgreen", size = 2) +
  scale_x_continuous(breaks = pc_data$PC_num) +
  labs(x = "Principal Component",
       y = "Cumulative Prop. Variance Explained") +
  theme_minimal()

# Combine two plots side-by-side using patchwork
combined_plot <- scree_plot + cum_plot

print(combined_plot)
```



```
# Save the combined plot to file
ggsave("../output/pca/pca_scree_and_cumulative_pve.png",
        plot = combined_plot, width = 12, height = 6, dpi = 300)
```

8. Interpretation

- The **loadings** (rotation) show how each original variable contributes to the PCs.
- The **principal component scores** ($\mathbf{x}$) are new features that can be used for clustering.
- From the **variance explained**, you can decide how many components to retain (e.g., enough to capture 90% of the variance).
- The **biplot** helps visualize the relationship between variables and observations in the reduced PC space.